

AMR 50 mm PVC VALVE ACTUATOR

Manufacturing Drawing, BOM, Assembly, and Prototype Release Pack

REVISION A-PROTOTYPE - NOT RELEASED FOR PRODUCTION OR SALE

- Motor: 5840-31ZY, 12 V, 40 rpm nominal, double shaft (real unit must be measured)
- Transmission: 40T/80T, module 1.5, PA20, 90.00 mm centre distance, 2:1 ratio
- Output: 20 mm shaft, dual 6204-2RS bearings, adjustable friction clutch, 30 mm manual hex
- Feedback: AS5600 on final shaft plus independent reeds and normally-closed hard limits
- Structure: 6 mm 6061-T6 plates plus 16 mm bearing carriers and replaceable valve clamp/adaptor
- Protection intent: IP65 minimum / IP67 preferred after production-intent ingress testing

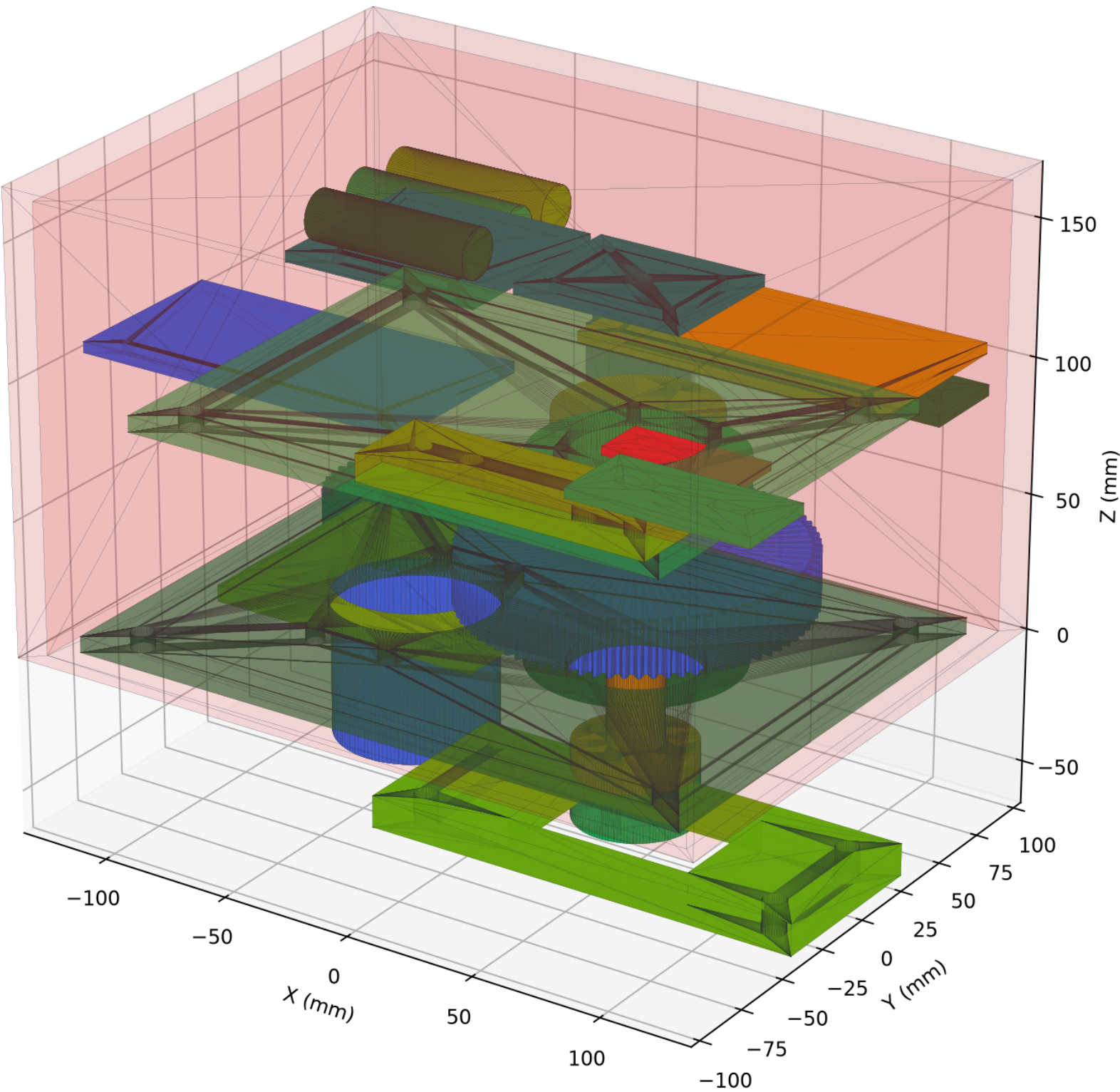
CRITICAL HOLD

Measure the actual motor, valve neck/stem, valve breakaway torque, and motor stall current before machining final interfaces.

The current repository TB6612 30 ms latching-valve firmware must not drive this continuous 12 V motor.

Resolved Assembly Envelope

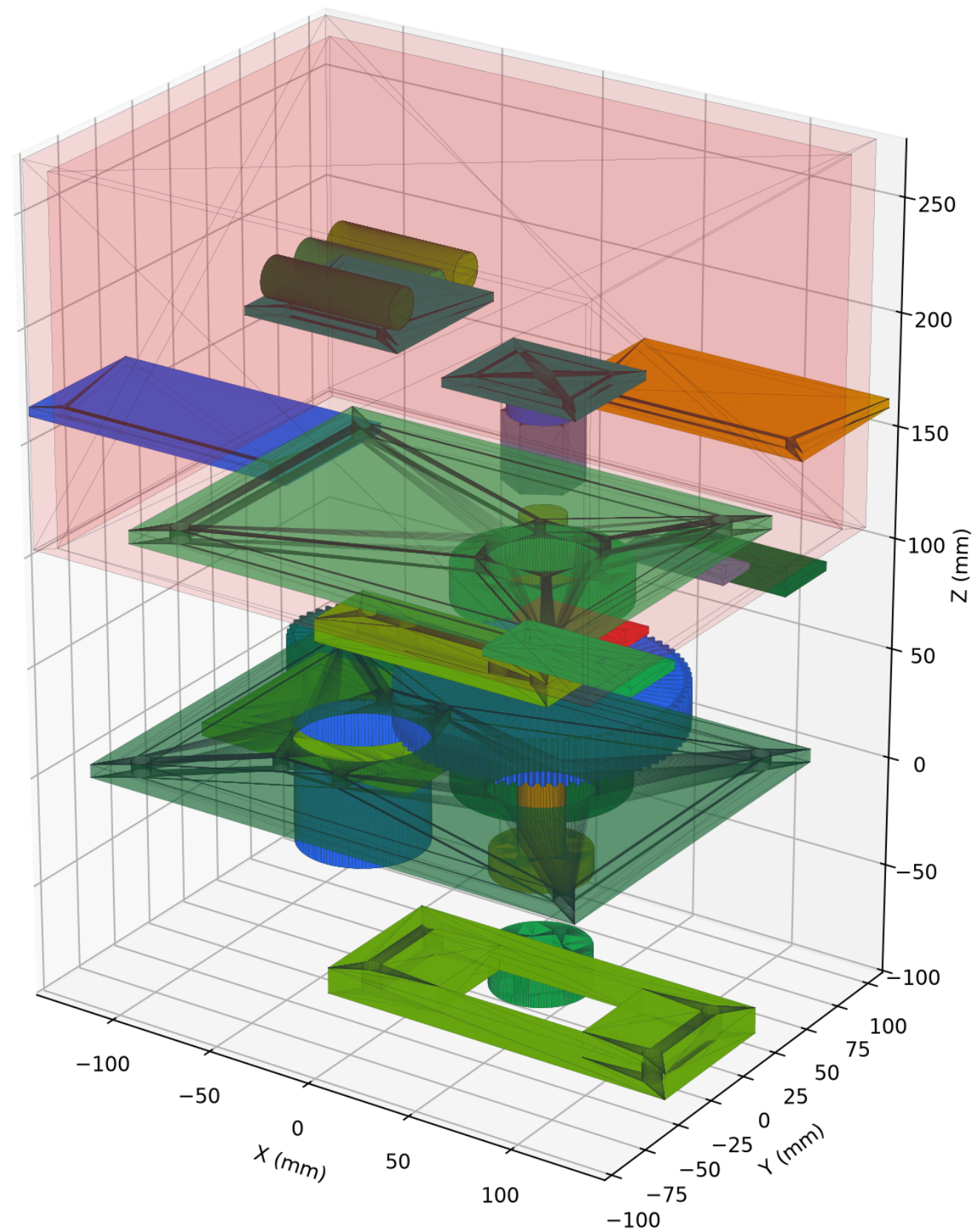
AMR 50 mm Valve Actuator - Assembly Envelope



Placed 50 mm configuration; purchased electronics are reference keep-out envelopes.

Exploded Assembly Arrangement

AMR 50 mm Valve Actuator - Exploded Arrangement



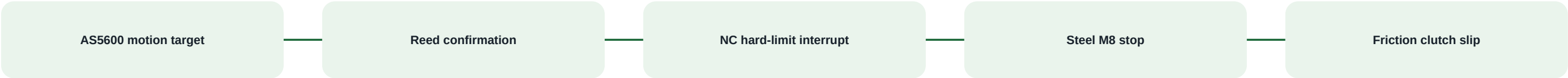
Exploded view is for component identification; use the written stack order for assembly.

Design Calculations and Protection Stack

Values are prototype sizing checks, not certification calculations.

Check	Equation / input	Result	Release interpretation
Gear ratio	80 / 40	2.00:1	Output speed 20 rpm nominal
Centre distance	$m(z1+z2)/2$	90.00 mm	Fixed datum; motor slots tune backlash
90 deg travel	0.25 rev / 20 rpm	0.75 s no-load	Use controlled PWM ramp; verify loaded time
Clutch preload	$T/(2 \mu r)$, T=8 Nm, μ =.30, r=25 mm	533 N	Select/test Belleville stack; wet friction matters
Gear tangential load	T/r , T=8 Nm, r=60 mm	133 N	200 N at 12 Nm before impact factor
20 mm shaft torsion	$16T/(\pi d^3)$, T=12 Nm	7.6 MPa	Key, flange, pin, and fatigue govern
Stop reaction	T/L , T=12 Nm, L=45 mm	267 N	Use $\geq 4\times$ impact factor for bracket check

Protection sequence



Final clutch torque is 1.5 times the highest measured valve breakaway torque only when that setting remains below the verified safe torque of the PVC stem, adapter bolts, printed gear teeth, motor gearbox, and stop structure. Measure dry/wet and hot/cold values; do not release using catalogue assumptions.

Mechanical BOM (1/2)

Item	Description	Qty	Material	Process	Critical note
P01	Lower base plate	1	6061-T6	Laser/CNC + anodize	240 x 180 x 6 mm
P02	Upper support plate	1	6061-T6	Laser/CNC + anodize	210 x 160 x 6 mm
P03	Adjustable motor bracket	1	6061-T6	Laser/CNC + anodize	+/-5 mm mesh adjustment
P04	40T motor gear	1	PA12-CF	FDM + metal D insert	m1.5, PA20, 17 mm face
P05	80T clutch gear	1	PA12-CF	FDM + face finish	m1.5, PA20, 20 mm face
P06	20 mm output shaft/flange	1	4140 or 316 SS	Turn/mill/grind	20 h6 journals; 4 x M6 flange
P07	Output gear clutch hub	1	C45 steel	Turn/keyway/zinc	OD50, bore20.1
P08	Clutch pressure plate	1	C45 steel	Turn/grind/zinc	OD62, bore17.2, t5
P09	Belleville stack	1	Spring steel	Purchased/calibrated	Initial preload about 530 N
P10	Replaceable stem adapter	1	17-4PH or PA12-CF	Machine/print	44 x 18; stem HOLD
P11	50 mm valve bridge clamp	1	6061-T6	CNC + anodize	72 mm HOLD opening
P12	63 mm alternative clamp	1	6061-T6	CNC + anodize	88 mm HOLD opening; alternative
P13	AS5600 bracket	1	6061 or PA12-CF	Machine/print	Nonmagnetic, adjustable
P14	Magnet carrier	1	PA12-CF	FDM	Final shaft mounting

Mechanical BOM (2/2)

Item	Description	Qty	Material	Process	Critical note
P15	Reed cam ring	1	PA12-CF	FDM	Two adjustable magnets
P16	OPEN reed bracket	1	316 SS	Laser	Slotted
P17	CLOSED reed bracket	1	316 SS	Laser	Slotted
P18	OPEN microswitch bracket	1	316 SS	Laser	+/-5 deg adjustment
P19	CLOSED microswitch bracket	1	316 SS	Laser	+/-5 deg adjustment
P20	Mechanical stop bracket	1	6061 or steel	Machine	Two M8 stops
P21	3S1P battery tray	1	ASA/PA12-CF	FDM	3 x protected 18650 envelope
P22	TTGO tray	1	ASA/PA12-CF	FDM	USB service access
P23	BTS7960 tray	1	ASA/PA12-CF	FDM	25 mm heatsink clearance
P24	Manual override hex	1	316 SS	Turn/mill/passivate	30 AF, bore20.2, cross-pin
P25	Gasketed enclosure cover	1	UV PC/ASA	FDM/mould	IP65/IP67 design intent
P26	Lower bearing carrier	1	6061-T6	Turn/mill/anodize	47 H7 x 16; retainers
P27	Upper bearing carrier	1	6061-T6	Turn/mill/anodize	47 H7 x 16; retainers

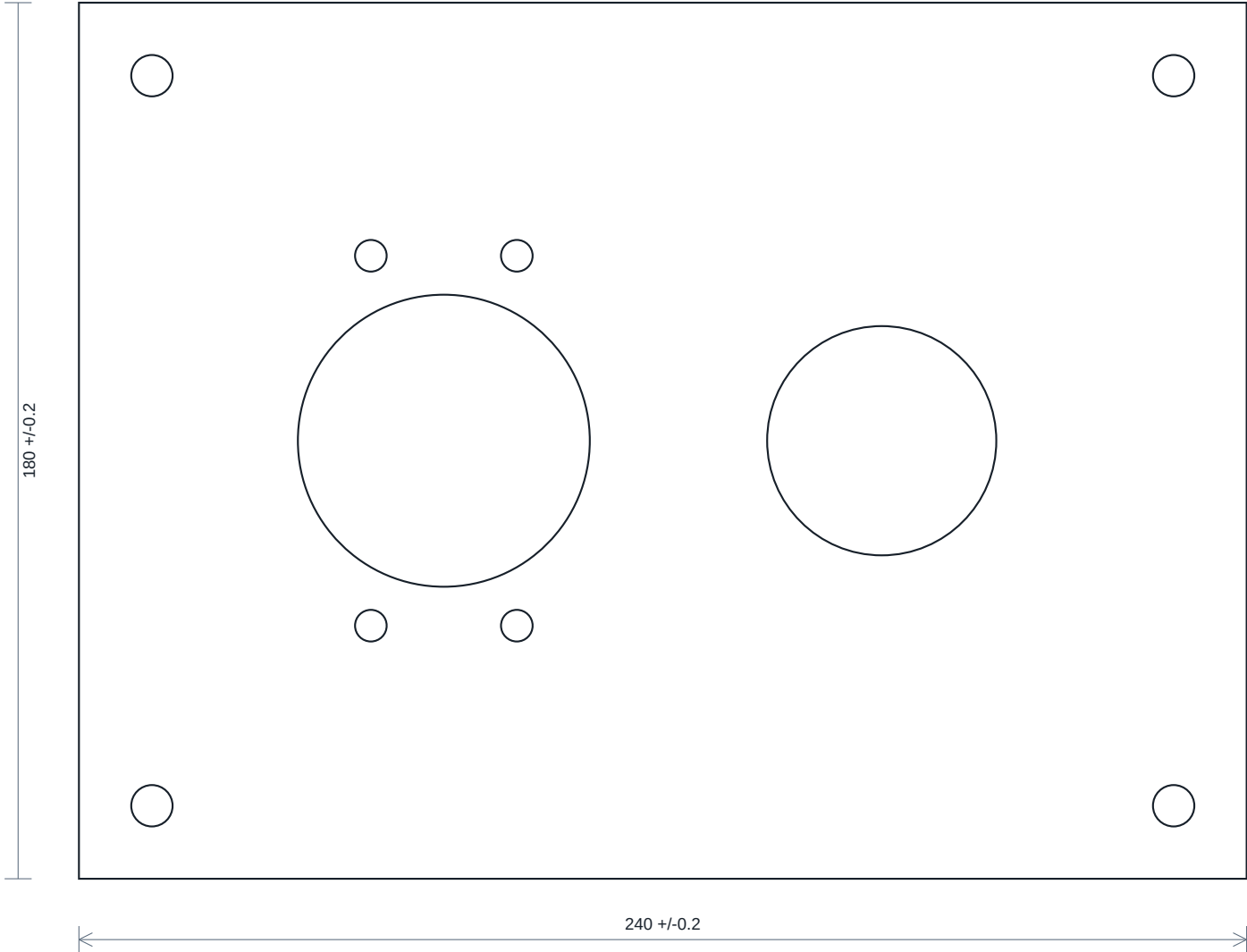
Material and Electrical BOM

Item	Description	Qty	Specification	Release note
M01	5840-31ZY double-shaft gearmotor	1	12 V, 40 rpm nominal	HOLD - measure supplier unit
M02	6204-2RS sealed bearing	2	20 x 47 x 14 mm	Known standard
M03	AS5600 angle module	1	3.3 V I2C	Measure PCB and magnet gap
M04	Diametric magnet	1	6 x 2.5 mm target	Match AS5600 vendor guidance
M05	Reed switch	2	Sealed dry contact	OPEN/CLOSED confirmation
M06	Reed cam magnet	2	Small NdFeB, retained	Pot and mechanically capture
M07	Long-lever microswitch	2	C/NO/NC, 250 VAC 5 A	Used at low-current control level
M08	Manual interlock switch	1	Two-pole NC + auxiliary	Hardware driver disable
M09	TTGO LoRa32 v2.1	1	868 MHz	Antenna separated from motor
M10	BTS7960 / IBT-2	1	High-current H bridge	Qualify against measured stall current
M11	MCP23017 input expander	1	3.3 V I2C	Reeds, limits, manual input
M12	3S BMS	1	Balancing + overcurrent	Rating from measured current
M13	Protected 18650 cell	3	Matched cells, 3S1P	12.6 V maximum
M14	Fuse and holder	1	DC rated	Size after motor-current test
M15	Logic DC/DC converter	1	12.6 V input to stable logic rail	Automotive/transient capable
M16	EPDM clamp liner	1	3 mm sheet	Protect PVC body
M17	EPDM enclosure gasket	1	Closed-cell	Compression verified by ingress test
M18	Cable glands	4	IP68, sized to cables	Strain relief
M19	Hydrophobic vent	1	IP67 pressure equalization	Above drainage plane
M20	Friction washers	2	Oil/water-stable friction material	Torque test wet and dry
M21	Bronze/steel gear sleeve	1	30.0 mm nominal interface	Final tolerance after wear test

Fastener BOM

Item	Description	Qty	Use
F01	M3 x 8 A4 socket screw	12	Electronics and sensor PCBs
F02	M3 A4 Nyloc + washer	12	Electronics
F03	M4 x 12 A4 socket screw	8	Sensor/reed brackets
F04	M4 A4 Nyloc + washer	8	Sensor/reed brackets
F05	M5 x 16 A4 socket screw	8	Microswitch and small structure
F06	M5 A4 Nyloc + washer	8	Small structure
F07	M6 x 20 A4 socket screw	16	Motor, carriers, shaft flange
F08	M6 A4 Nyloc + washer	16	Motor, carriers, shaft flange
F09	M8 x 70 A4 bolt	4	Valve bridge clamp; confirm length
F10	M8 A4 Nyloc + large washer	4	Valve bridge clamp
F11	M8 x 35 steel stop screw	2	Mechanical stops
F12	M8 jam nut	4	Two per stop
F13	M16 prevailing-torque nut	1	Clutch preload
F14	M16 Belleville washers	1	Select stack for about 530 N initial preload
F15	6 x 6 mm parallel key	1	Output hub; length per hub
F16	5 mm stainless cross-pin	1	Manual override
F17	47 mm internal bearing retainer	4	Two per bearing carrier or drawing-equivalent plates
F18	M4 A4 enclosure screw	12	Gasketed cover; captive preferred

P01 Lower Base Plate



Bearing-carrier locator: DIA47.1 at X=+45, Y=0

Corner holes: 4 x DIA8.5 at X=+/-105, Y=+/-75

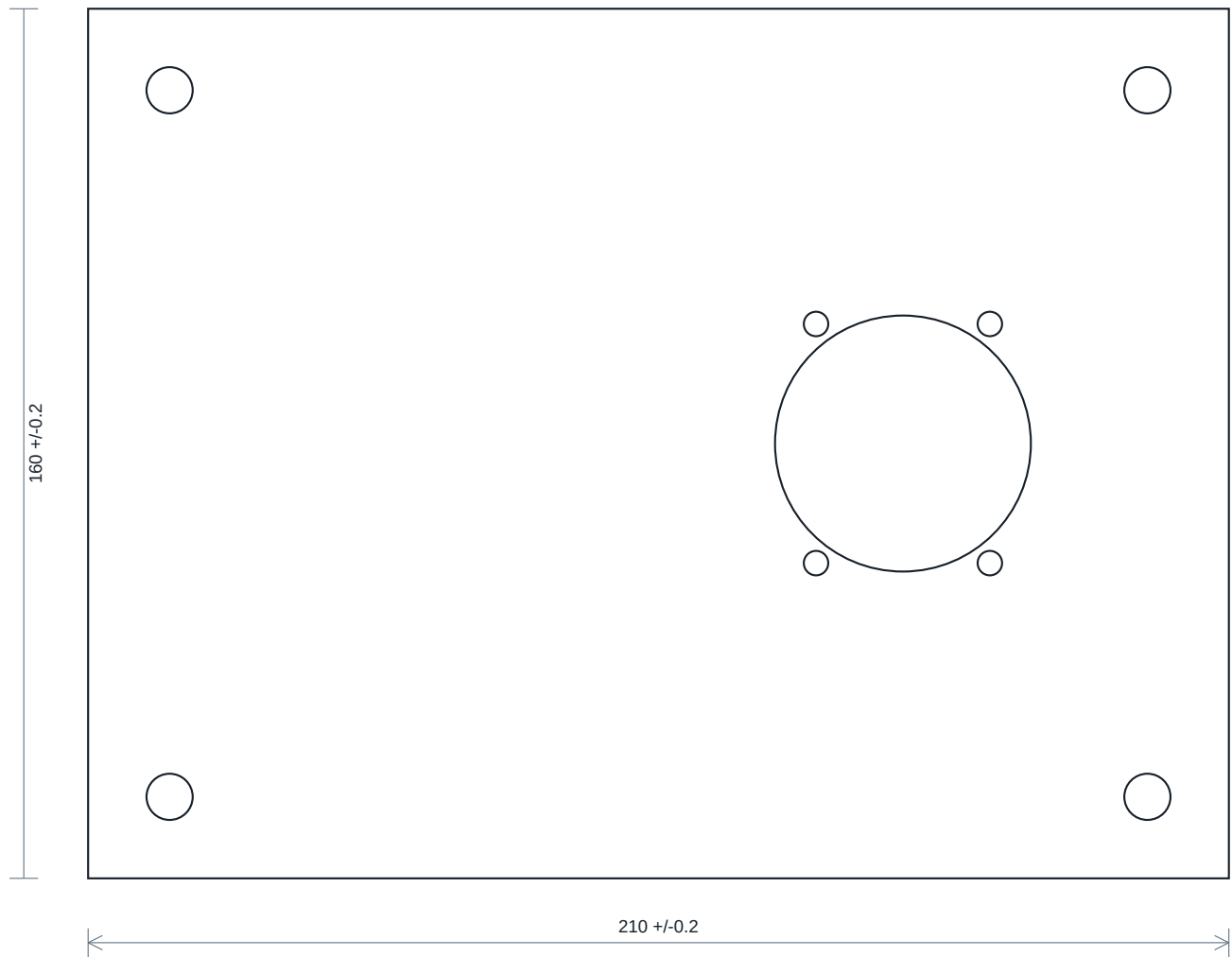
Motor clearance: DIA60 at X=-45; 4 x DIA6.5 slots/holes

MANUFACTURING NOTES

- 1. Plate is not the bearing seat; bolt P26 carrier to this plate.
- 2. Maintain output/motor gear datum centres at 90.00 mm.
- 3. Flatness 0.3 mm over plate; break all edges 0.5 mm max.

P01 LOWER BASE PLATE	REV: A-PROTOTYPE
Material: 6061-T6, 6 mm	QTY: 1 UNITS: mm
Finish: Clear anodize; mask interfaces	GENERAL: ISO 2768-m; deburr

P02 Upper Support Plate



Carrier locator: DIA47.1 at X=+45, Y=0

Structure holes: 4 x DIA8.5 at X=+/-90, Y=+/-65

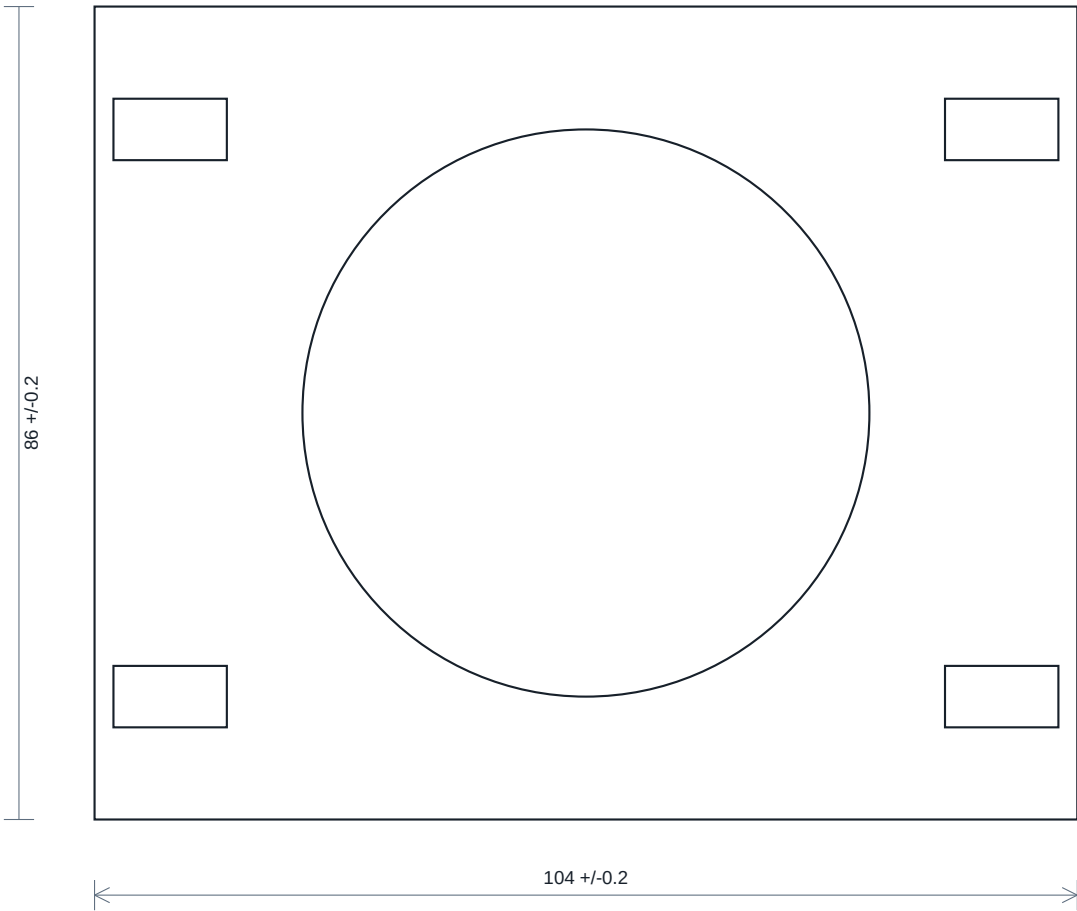
AS5600 holes: 4 x DIA4.5 about output axis

MANUFACTURING NOTES

- 1. P27 bearing carrier mounts below this plate.
- 2. AS5600 support must be nonmagnetic and vertically adjustable.
- 3. Flatness 0.3 mm over plate.

P02 UPPER SUPPORT PLATE	REV: A-PROTOTYPE
Material: 6061-T6, 6 mm	QTY: 1 UNITS: mm
Finish: Clear anodize; mask interfaces	GENERAL: ISO 2768-m; deburr

P03 Adjustable Motor Bracket



4 slots: 12 x 6.5

Slot centres: X= $\pm$ 44, Y= $\pm$ 30

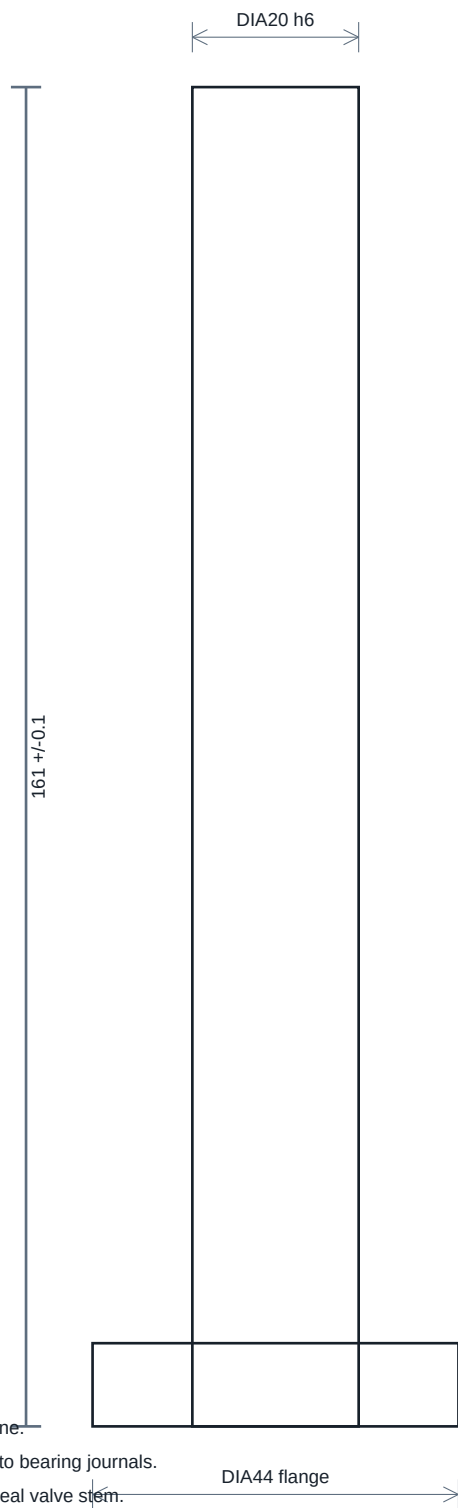
Centre/body clearance: DIA60

MANUFACTURING NOTES

- 1. Slots provide approximately +/-5 mm backlash adjustment.
- 2. Final motor mount holes/cradle walls depend on measured supplier unit.
- 3. Set final shaft centre exactly 90.00 mm from output axis.

P03 MOTOR BRACKET	REV: A-PROTOTYPE
Material: 6061-T6, 6 mm	QTY: 1 UNITS: mm
Finish: Clear anodize	GENERAL: ISO 2768-m; deburr

P06 Output Shaft and Adapter Flange



MANUFACTURING NOTES

- 1. Do not transfer torque by set screws alone.
- 2. Maintain flange face runout <=0.05 mm to bearing journals.
- 3. Confirm 1-2 mm axial clearance above real valve stem.

Flange: t10, 4 x DIA6.6 on PCD30

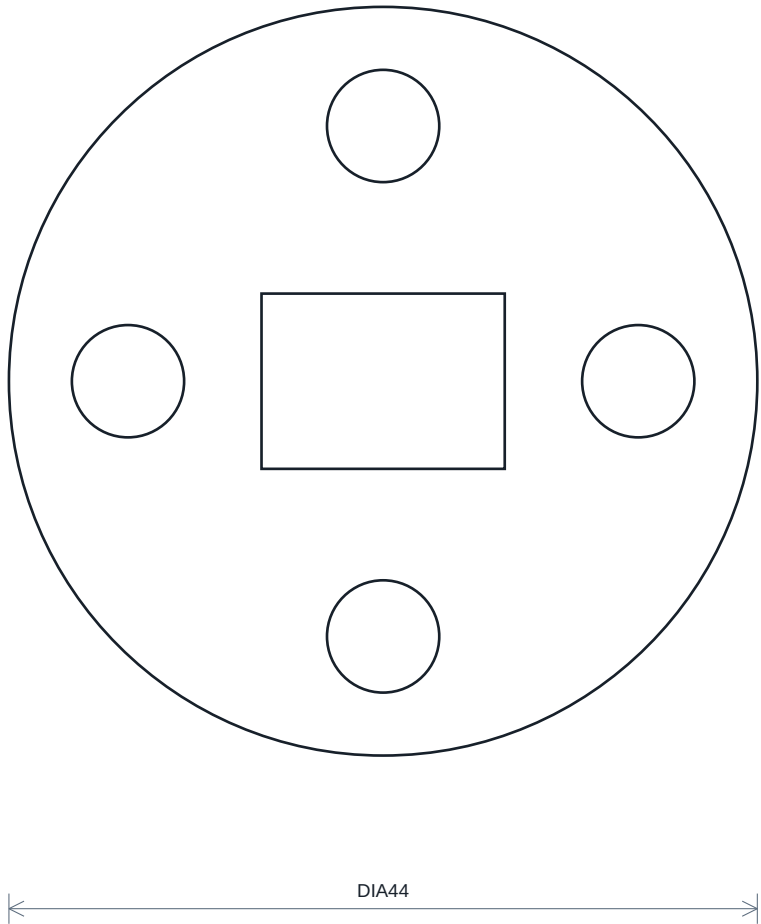
Bearing journals: DIA20 h6; Ra <=0.8 um

Keyway: 6 mm, machine to hub/key standard

Manual cross-pin: DIA5 H11, locate after stack measurement

P06 OUTPUT SHAFT AND FLANGE	REV: A-PROTOTYPE
Material: 4140 steel or 316 SS	QTY: 1 UNITS: mm
Finish: Grind/passivate or zinc-nickel	GENERAL: ISO 2768-m; deburr

P10 Replaceable Valve Stem Adapter



Thickness 18 +/-0.1

4 x DIA6.6 on PCD30

Prototype socket: 14.3 x 10.3 THROUGH

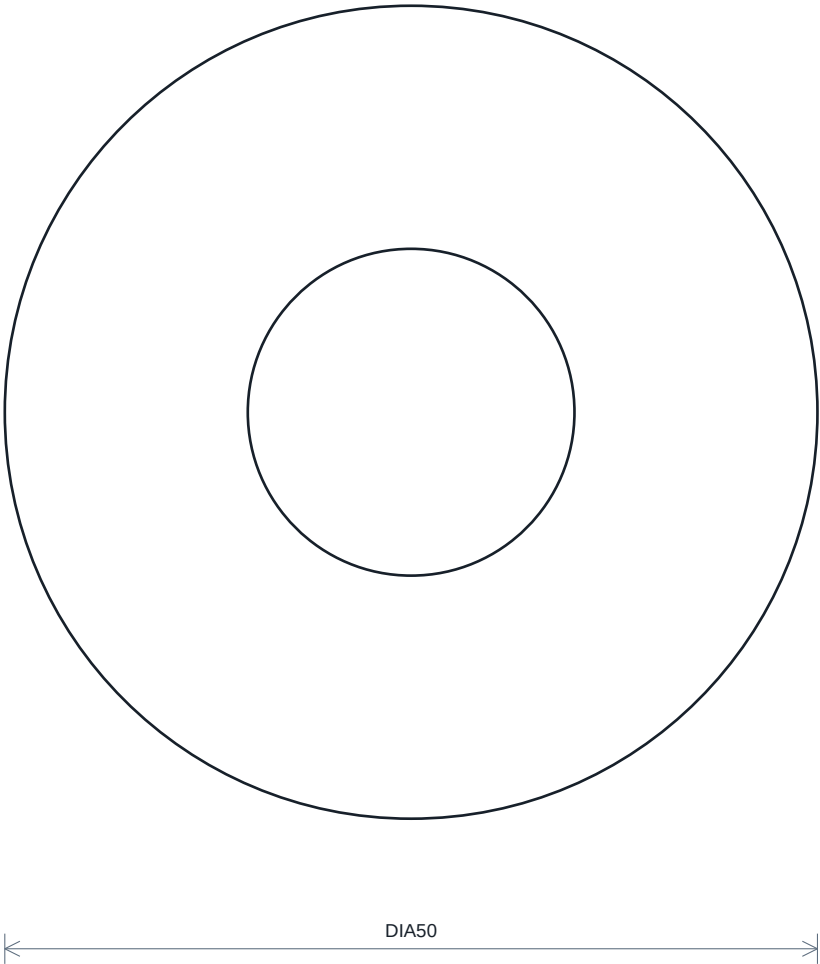
FINAL SOCKET: HOLD - MEASURE REAL STEM

MANUFACTURING NOTES

- 1. Use a new insert for each valve stem family.
- 2. Provide 1-2 mm axial clearance; never preload PVC stem axially.
- 3. Bolt to P06 flange with four locked M6 fasteners.

P10 VALVE STEM ADAPTER	REV: A-PROTOTYPE
Material: 17-4PH or reinforced polymer	QTY: 1 UNITS: mm
Finish: Passivate if metal	GENERAL: ISO 2768-m; deburr

P07 Clutch Output Hub



Sleeve height 26; flange height 6

Bore DIA20.1 before keyway

Keyway: 6 mm to selected shaft fit

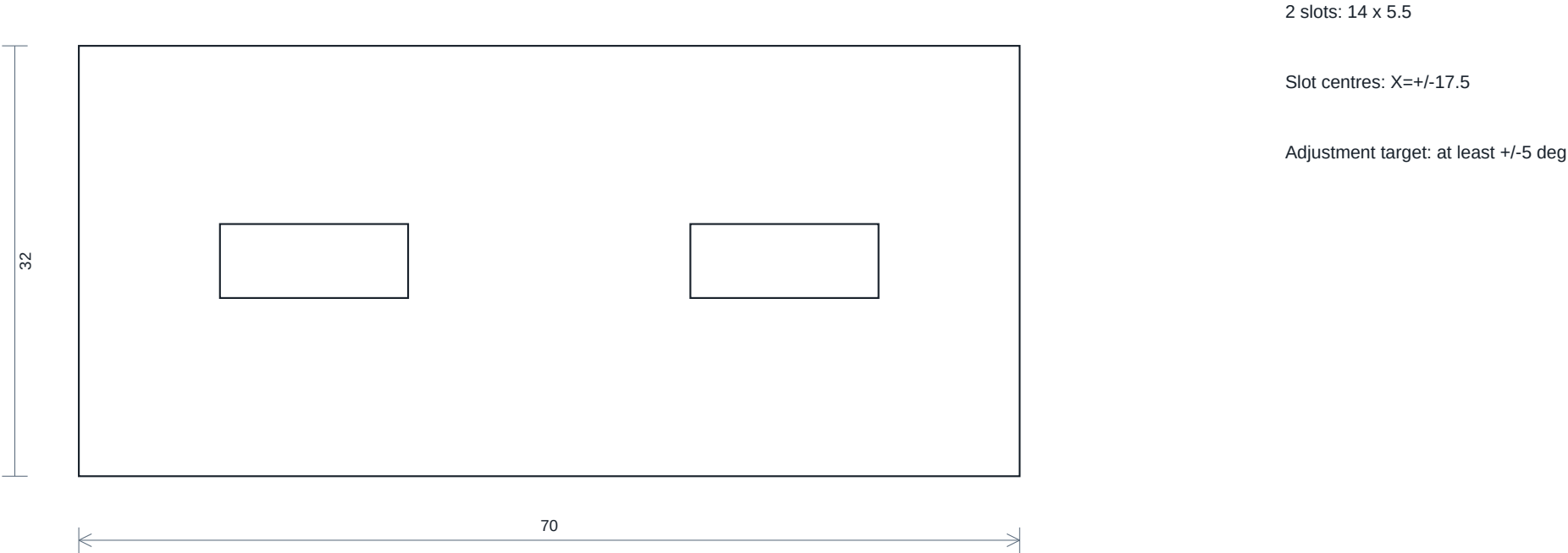
Sleeve DIA30; friction face Ra 1.6-3.2 um

MANUFACTURING NOTES

- 1. Use positive key or cross-bolt; set screws are retention only.
- 2. Keep zinc and grease off friction face.
- 3. Verify sleeve permits free gear rotation before clutch preload.

P07 CLUTCH OUTPUT HUB	REV: A-PROTOTYPE
Material: C45 steel	QTY: 1 UNITS: mm
Finish: Zinc-nickel; mask friction/key surfaces	GENERAL: ISO 2768-m; deburr

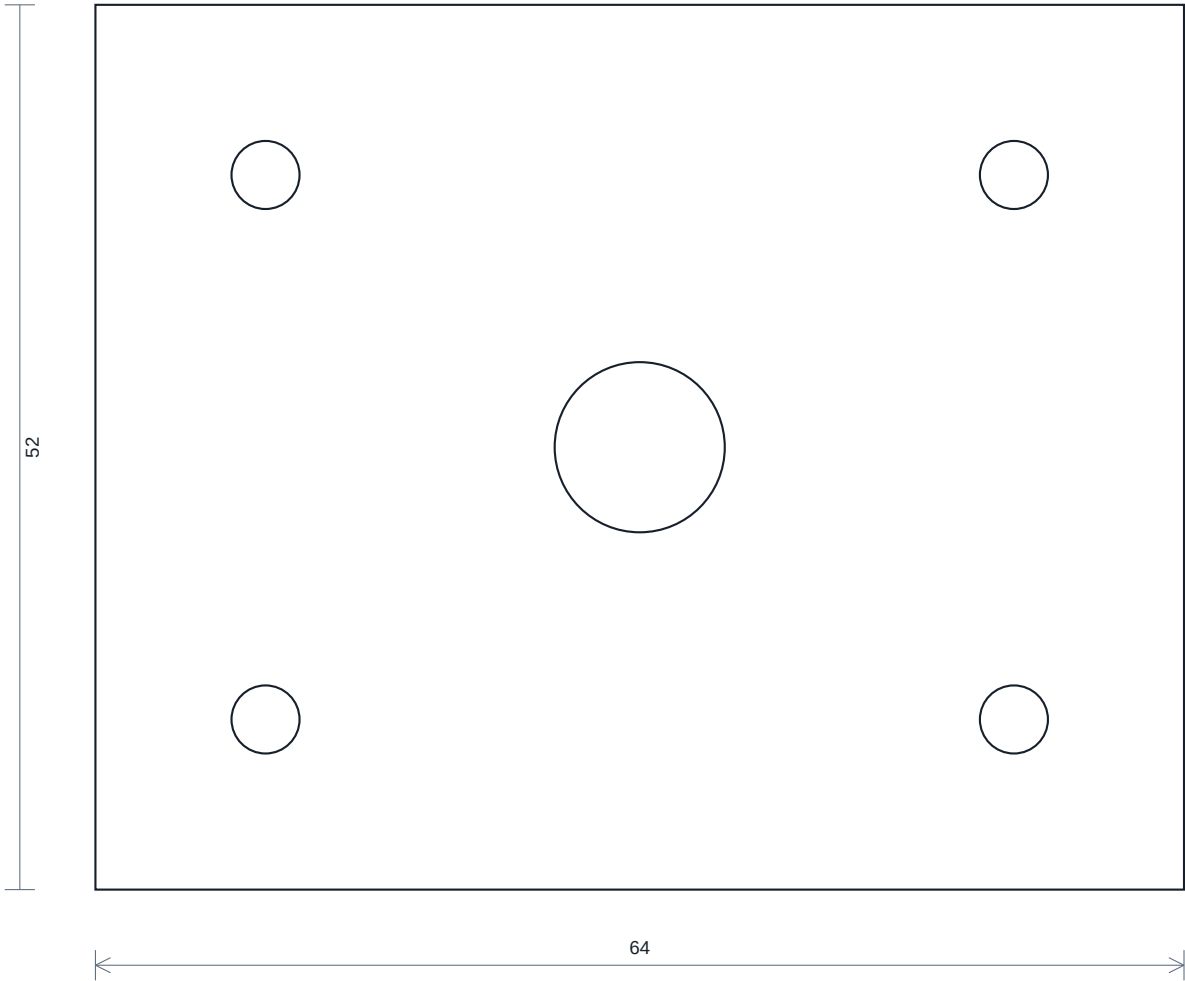
P18/P19 OPEN and CLOSED Microswitch Brackets



- MANUFACTURING NOTES**
- 1. Use long-lever C/NO/NC microswitches.
 - 2. Wire NC contact to interrupt hazardous motor direction in hardware.
 - 3. Mount so reverse motion remains possible at either limit.

P18/P19 LIMIT SWITCH BRACKETS	REV: A-PROTOTYPE
Material: 316 stainless, 4 mm	QTY: 2 UNITS: mm
Finish: Deburr/passivate	GENERAL: ISO 2768-m; deburr

P13 AS5600 Adjustable Bracket



Centre clearance DIA10

4 x DIA4 at X= $\pm$ 22, Y= $\pm$ 16

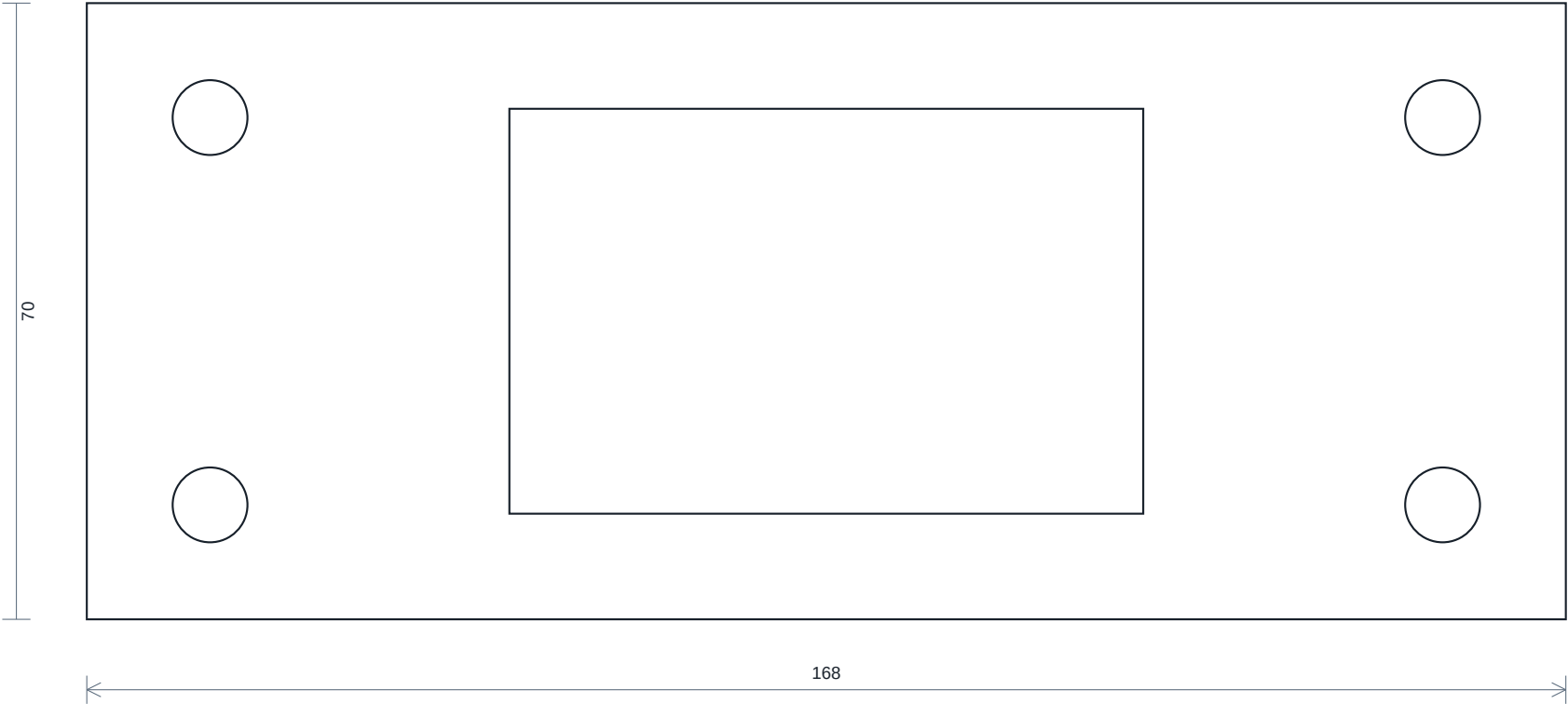
Add vertical slots to suit measured module/gap

MANUFACTURING NOTES

- 1. Magnet must be diametrically magnetized and concentric with final shaft.
- 2. Use nonmagnetic screws/support near sensor.
- 3. Calibrate actual valve CLOSED/OPEN angles after assembly.

P13 AS5600 BRACKET	REV: A-PROTOTYPE
Material: 6061 or PA12-CF, 4 mm	QTY: 1 UNITS: mm
Finish: Nonmagnetic	GENERAL: ISO 2768-m; deburr

P11 50 mm Valve Bridge Clamp



Opening: 72 x 46 HOLD

4 x DIA8.5 at X= $\pm$ 70, Y= $\pm$ 22

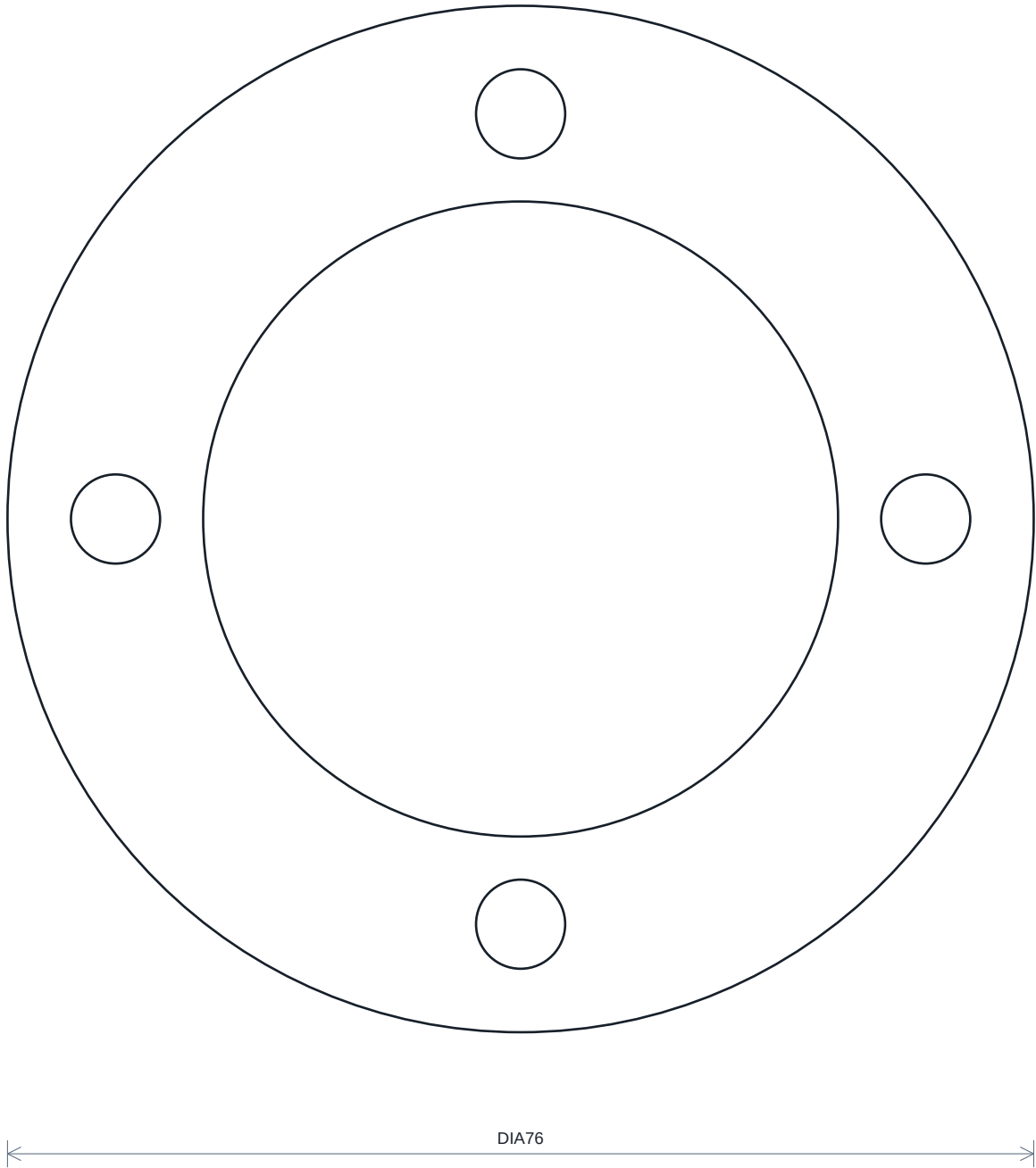
Use 3 mm EPDM liner

MANUFACTURING NOTES

- 1. Machine opening only after measuring real valve neck/body landing.
- 2. Clamp the strong valve body/neck; never react torque through thin pipe.
- 3. P12 changes opening to 88 mm for the 63 mm placeholder configuration.

P11 50 mm VALVE BRIDGE CLAMP	REV: A-PROTOTYPE
Material: 6061-T6, 12 mm	QTY: 1 UNITS: mm
Finish: Clear anodize + 3 mm EPDM	GENERAL: ISO 2768-m; deburr

P26/P27 6204 Bearing Carriers



Bearing bore: DIA47 H7

Width: 16 +/-0.05

4 x DIA6.6 on PCD60

Internal retainers both sides

MANUFACTURING NOTES

- 1. Finish-machine bearing bore after anodize allowance.
- 2. Use force on bearing outer race only.
- 3. Carriers, not 6 mm plates, provide full 14 mm bearing support.

P26/P27 BEARING CARRIERS	REV: A-PROTOTYPE
Material: 6061-T6, 16 mm	QTY: 2 UNITS: mm
Finish: Clear anodize; mask bore	GENERAL: ISO 2768-m; deburr

Assembly and Calibration Sequence

- 1 Measure motor and valve; machine HOLD interfaces only after inspection approval.
- 2 Install 6204 bearings in P26/P27 carriers and bolt carriers to P01/P02.
- 3 Install P06 shaft, P10 measured adapter, valve clamp, and verify 1-2 mm axial clearance.
- 4 Install motor and 40T/80T gears at 90.00 mm centres; tune 0.25-0.35 mm backlash.
- 5 Assemble clutch: hub, friction washer, gear, friction washer, pressure plate, Belleville stack, M16 locknut.
- 6 Install AS5600, reed cams, hard limits, steel stops, and manual hardware interlock.
- 7 Calibrate clutch from 8 N m and set final torque from measured valve data.
- 8 Bench test current-limited, then complete ingress, fault, temperature, radio, and endurance tests.

DO NOT CONNECT TO CURRENT TB6612 PULSE FIRMWARE

This actuator requires continuous bidirectional PWM, AS5600 feedback, independent reeds/limits, manual inhibit, movement timeout, and current/stall protection.

Both hard limits and manual mode must remove hazardous motor drive through hardware, even if the MCU or software fails.

Prototype Release Hold Points

Status	Hold point	Evidence required
OPEN	Motor interface	Body, mount, shaft, D-flat, projection, stall current at 9.0/11.1/12.6 V
OPEN	Valve interface	Neck/body, stem profile/height, axial float, breakaway and running torque wet/dry/hot/cold
OPEN	Power train	Gear contact, backlash, key/pin/flange, clutch wet/dry torque, stop impact load
OPEN	Electronics	Driver/BMS/fuse/wire ratings, surge/ESD/reverse polarity, brownout and thermal behavior
OPEN	Functional safety	Two hard limits, manual inhibit, AS5600 faults, no motion, reversed motion, timeout, radio loss
OPEN	Environment	IP65/IP67, condensation, UV, corrosion, vibration, temperature, EMC and LoRa coexistence
OPEN	Durability	500 engineering cycles and 5,000 production-intent pilot cycles
OPEN	Commercial	Risk assessment, labels, traceability, instructions, maintenance, warranty, Tunisian compliance review